

452 George Street

Commercial Fitout — Levels 1 to 4

HYDRAULIC SERVICES — DESIGN CALCULATIONS

Cold Water Distribution System

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Issue	B — For Construction
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Client	Meridian Property Group
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1. Document Control

Rev	Date	Description	Prepared	Checked	Approved
A	14 Apr 2026	Preliminary Issue — For Review	JH	SP	—
B	09 May 2026	Revised — For Construction (pipe schedule updated, pump duty confirmed)	JH	SP	AW

2. Project Description

The project involves a full commercial fitout of Levels 1 to 4 at 452 George Street, Sydney CBD. The building is a reinforced concrete structure constructed circa 1988. The existing hydraulic services are to be decommissioned at each tenancy level and replaced with new cold water, hot water, sanitary, and drainage services to suit the proposed fitout configuration.

These calculations address the cold water distribution system only. Hot water, sanitary plumbing, drainage, and stormwater calculations are contained in separate calculation packages (BCA-HYD-CAL-002 through BCA-HYD-CAL-004).

3. Design Basis

Building type	Class 5 — Commercial Office
Levels served	Levels 1, 2, 3, and 4 (Ground Floor existing, not in scope)
Occupancy	Approximately 320 persons across all tenancy levels
Fitout type	Open-plan offices with central amenities cores
Water supply	Sydney Water reticulation — George Street main
Available mains pressure	350 kPa (static, confirmed per Sydney Water pressure enquiry Ref: SW-ENQ-2026-04471, dated 28 March 2026)
Minimum fixture pressure	Refer Section 5 — AS/NZS 3500.1:2021 Cl. 3.3.3
Design standard	AS/NZS 3500.1:2021 — Water Services
Pipe material	Copper tube — Type B to AS 1432, above ground HDPE SDR 11 to AS/NZS 4130, below ground
Pipe sizing method	Velocity + pressure-drop iterative method Maximum velocity: 3.0 m/s (AS/NZS 3500.1 Cl. 3.4) Minimum velocity: 0.5 m/s (self-cleansing)
Hazen-Williams C	Copper (new): C = 130
Diversity factor	0.65 — commercial office, multi-tenancy (refer Table 3.8.3, AHSCA Guide)
Floor-to-floor height	3,900 mm (structural), 400 mm slab — net 3,500 mm per floor
Riser offset	No offset — riser routes vertically from B1 plant room

4. Referenced Standards and Documents

Reference	Title
AS/NZS 3500.1:2021	Plumbing and drainage — Water services

AS 1432:2004	Copper tubes for plumbing, gasfitting and drainage applications
AS/NZS 4130:2009	Polyethylene (PE) pipes for pressure applications
NCC 2022 Vol. 1	National Construction Code — Performance Requirements
AHSCA Guide Ed. 4	Association of Hydraulic Services Consultants Australia — Design Guide
Sydney Water Ref. SW-ENQ-2026-04471	Pressure and flow enquiry — 452 George Street
Arch. Dwg. BCA-A-100 Rev C	Floor plans — Levels 1–4
Structural Dwg. BCA-S-200 Rev B	Services penetration schedule

5. Fixture Loading and Flow Rate Derivation

Fixture units (FU) have been assigned per AS/NZS 3500.1:2021 Table 3.8.1. The simultaneous demand flow rate Q has been derived using the diversity factor method with a commercial office diversity factor of 0.65. Each amenities core serves one tenancy level.

5.1 Per-Level Fixture Schedule

Fixture Type	No. Off (per level)	FU Each (AS 3500.1 Tbl 3.8.1)	FU Total (per level)	Cold Water FU ($\times 0.75$)	Design Flow Q (L/s)
WC — Cistern flush	10	2.0	20.0	15.0	—
Hand basin — tap	8	1.0	8.0	6.0	—
Kitchen sink — single	4	2.0	8.0	6.0	—
Cleaners sink	2	3.0	6.0	4.5	—
Drinking fountain	2	0.5	1.0	0.75	—
TOTAL per level	26	—	43.0	32.25	$0.65 \times Q_{\text{peak}}$

5.2 Simultaneous Demand — All Levels

The design flow rate is derived from the total fixture unit loading across all four levels, applying the diversity factor from AS/NZS 3500.1:2021. Total cold water FU (4 levels): $32.25 \times 4 = 129.0$ FU. From AS/NZS 3500.1 Table 3.8.2, peak flow at 129 FU ≈ 4.80 L/s. Applied diversity factor 0.65: $Q_{\text{design}} = 4.80 \times 0.65 = \mathbf{3.12 \text{ L/s}}$ (demand at base of riser, serving all four levels simultaneously).

6. Available Pressure Analysis

Pressure is analysed along the critical path from the Sydney Water connection point at the ground floor plant room (Elevation ± 0.000) to the most hydraulically remote fixture, identified as the Level 4 north-east corner hand basin (Elevation $+14.400$ m above ground floor slab).

6.1 Static Pressure Budget

Available static pressure at point of connection	+350.0 kPa
Less: static head loss — elevation gain 14.40 m ($14.40 \times 9.81 \times 1.0 \text{ kN/m}^3 \div 1000$)	– 141.3 kPa
Less: water meter pressure loss (Sydney Water DN50 meter at 3.12 L/s)	– 18.5 kPa
Less: backflow prevention device (Watts 909 reduced-pressure zone device, DN50, at 3.12 L/s)	– 28.0 kPa
Less: isolating valve losses ($\times 3$ gate valves)	– 3.5 kPa
Less: friction losses — distribution pipework (cold water ring main + branch risers, see Section 7)	– 97.6 kPa
= Residual pressure at remote fixture	= 61.1 kPa
Minimum required at fixture (AS/NZS 3500.1 Cl. 3.3.3 — hand basin tap)	50.0 kPa
MARGIN	+ 11.1 kPa

Note: Residual margin of 11.1 kPa is positive but narrow. Any increase in friction losses or reduction in mains pressure below 350 kPa will result in sub-minimum pressure at the remote fixture. A pressure booster pump should be considered if mains pressure is confirmed below 350 kPa on site.

7. Cold Water Pipe Schedule and Velocity Calculations

The following pipe schedule sets out each pipe segment from the point of connection to the most hydraulically remote fixture. Design flows are derived from Section 5. Velocities and pressure drops are calculated using the Hazen-Williams equation ($C = 130$ for new copper) and the continuity equation $Q = V \cdot A$. All pipe sizes are nominal DN (inside diameter per AS 1432 Type B). Segments highlighted red indicate values that require review.

Velocity limit: 3.0 m/s maximum (AS/NZS 3500.1:2021 Cl. 3.4) | **Minimum velocity:** 0.5 m/s | **Hazen-Williams C:** 130 (copper, new)

Seg. Ref	Pipe Segment Description	Nom. DN	Int. Dia. (mm)	Length (m)	Design Flow Q (L/s)	Velocity V (m/s) CALC	Stated V (m/s) DOC	$\Delta p/m$ (kPa/m)	ΔP seg. (kPa)	Status
CW-01	Point of connection to riser base (Ground floor plant room)	DN65	62.0	8.0	3.12	1.03	1.03	0.221	1.8	✓
CW-02	Main riser — Ground floor to Level 2 (Vertical run, incl. 2× 90° elbows)	DN50	47.0	8.2	3.12	1.80	1.80	0.853	7.0	✓
CW-03	Main riser — Level 2 to Level 4 (Vertical run, incl. 2× 90° elbows)	DN50	47.0	8.2	2.08	1.20	1.20	1.512	12.4	✓
CW-04	Level 1–2 branch to amenities core (Horizontal ring main, Level 2)	DN20	18.0	22.5	1.04	4.09	2.51	11.945	268.8	■ SEE NOTE
CW-05	Level 3 branch — riser to amenities core (Horizontal distribution)	DN32	29.5	19.0	1.04	1.52	1.52	1.077	20.5	✓
CW-06	Level 4 branch — riser to amenities core (Horizontal distribution)	DN25	23.0	19.0	0.52	1.25	0.87	1.003	19.1	■ SEE NOTE
CW-07	Level 4 sub-branch — WC cisterns (5 off) (DN20 branch from ring main)	DN20	18.0	12.0	0.42	1.65	1.65	2.228	26.7	✓
CW-08	Level 4 sub-branch — basins and sinks (Remote branch to NE corner)	DN15	13.0	11.5	0.26	1.96	1.96	4.471	51.4	✓
TOTAL FRICTION LOSS — CRITICAL PATH									407.7	

- Table Notes:**
- (1) Column "Velocity V (m/s) CALC" shows velocity calculated from $Q = V \cdot A$ using stated DN and internal diameter. Column "Stated V (m/s) DOC" shows the velocity recorded in the designer's working calculations. Discrepancies between these columns indicate a calculation error.
 - (2) $\Delta p/m$ and ΔP calculated using Hazen-Williams: $h_f/L = (10.67 \times Q^{1.852}) / (C^{1.852} \times D^{4.87})$, converted to kPa. $C = 130$ throughout.
 - (3) Internal diameters per AS 1432:2004 Table 1 — Copper tube, Type B.
 - (4) Highlighted segments require review prior to construction issue — refer Section 8.

8. Calculation Review — Discrepancy Register

The following discrepancies were identified during internal checking (Issue B). Each item must be resolved and verified by the design engineer prior to construction. This register forms part of the document audit trail.

Segment	Issue	Stated Value	Calculated Value	Discrepancy
CW-04	Velocity exceeds AS/NZS 3500.1 Cl. 3.4 limit	V = 2.51 m/s (DN25 assumption)	V = 4.09 m/s (DN20, ID=18.0mm)	Pipe drawn as DN20 but sized as DN25 — upsize to DN25 required
CW-06	Stated velocity does not reconcile with Q and internal diameter	V = 0.87 m/s	V = 1.25 m/s (Q=0.52 L/s, ID=23.0mm)	Likely used OD (28mm) instead of ID (23mm) in area calculation
CW-03	Pressure drop understated — affects residual pressure budget	$\Delta P = 12.4 \text{ kPa}$	$\Delta P = 3.3 \text{ kPa}$ (Hazen-Williams, C=130)	Total friction loss understated by -9.1 kPa. Revised residual margin = 20.2 kPa (positive)

9. Pump Requirement Assessment

Based on the pressure analysis in Section 6, no pressure booster pump is required provided mains static pressure is confirmed at or above 350 kPa at the point of connection. However, resolution of the CW-03 pressure drop discrepancy (Section 8) reduces the calculated residual margin. The designer should confirm whether a booster pump duty is required following correction of the pipe schedule.

Booster pump required?	To be confirmed — subject to CW-03 resolution (see Section 8)
If required, duty point	Q = 3.12 L/s at H = 30 kPa minimum
Standby arrangement	Duty/standby configuration recommended
Set point pressure	400 kPa downstream of pump (if installed)

10. Compliance Statement

Subject to resolution of the discrepancies recorded in Section 8, the cold water distribution system for 452 George Street has been designed in general accordance with AS/NZS 3500.1:2021 and the design criteria set out in Section 3 of this document. This calculation package does not constitute approval under the National Construction Code or any local authority. Formal plumbing approval is to be obtained by the contractor prior to commencement of installation works.

END OF DOCUMENT

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